

A Appendix

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A.1 Premise-Hypothesis Pairs & Few-Shot Learning

This section explains how I construct the IP evaluation score from NTE texts using a few-shot, premise-hypothesis approach. Few-shot learning means adapting a large, already pre-trained language model to a new task with a small, hand-labeled set of examples instead of a large training corpus. It leverages the model’s prior linguistic knowledge and is well suited when expert labels are costly. I frame stance detection as natural language inference because it forces the model to answer a concrete question about each passage: does this text “support”, “contradict”, or “say nothing” about a specific claim? Concretely, the premises are paragraph-level excerpts from the NTE reports, and the hypotheses are short, standardized statements that encode evaluation claims, e.g, “USTR believes that there is widespread IP violation in the country.” For each premise-hypothesis pair, the model returns probabilities for “entailment”. In other words, the model assesses whether a premise (IP paragraph from NTE) supports, or “entails”, a hypothesis. I convert these to a signed stance value (support for a negative IP claim moves the score downward; support for a positive claim moves it upward; neutral stays near zero), then aggregate within country-year across all pairs.

I employ multiple hypotheses to build this new IP evaluation proxy primarily because IP protection is a multidimensional concept that involves distinct factors including enforcement efforts and legal provisions. Each IP paragraph from a country’s NTE report forms a single premise and is paired with each hypothesis that appears in Table A.1, which allows the DeBERTa model to compute the probability of the premise entailing the hypothesis, namely entailment probabilities. The final score is derived by computing a weighted sum of the model’s entailment probabilities across multiple hypotheses and then scaling it to the -5 to $+5$ range.

To minimize researcher discretion while capturing variation in institutional signal strength, I assign differential weights only to text segments tied to Special 301 Watch List designations (Watch List, Priority Watch List, or Priority Foreign Country) and treat all other segments with unit weight. The rationale is informational, not theoretical: Watch List placements follow a formal USTR process and constitute an adjudicated, high-salience

Table A.1: Hypotheses & weights

Hypothesis	Weight
H1: The country is the Priority Foreign Country.	−2
H2: The country is on the Priority Watch List.	−2
H3: The country is on the Watch List.	−1.5
H4: The country has markets listed as the Notorious Market.	−1.5
H5: The author of this text believes that the country does not put in efforts to combat IP violations.	−1
H6: The author of this text believes that the country has made efforts to combat IP violations.	+1
H7: The author of this text supports the passage of the new IP legislation in the country.	+1
H8: The author of this text opposes the passage of the new IP legislation in the country.	−1
H9: The author of this text believes that there is widespread IP violation in the country.	−1.5
H10: The author of this text believes that the country is lack of resources to combat IP violations.	−1
H11: The author of this text believes that the country has strong IP law.	+2
H12: This text mentions the increase of IP violations in the country.	−1
H13: This text mentions the decrease of IP violations in the country.	+1

Note: All 13 hypotheses are used for both training and scoring.

statement of US dissatisfaction with a partner’s IP regime. Accordingly, when aggregating paragraph-level stance scores into the country–year measure, segments that explicitly reference a Watch List placement receive a differential weight, while all other segments receive indifferent unit weights, which yields a weighted average that reflects the greater evidentiary content of these Watch List classifications. It is important to note that the weights are not tuned to the outcome but are set *ex ante* and applied uniformly.

I compile a set of 300 hand-labeled NTE IP paragraphs, each paired with all 13 hypotheses to produce up to 3,900 premise–hypothesis NLI pairs. Each pair is labeled to indicate whether the text does or does not entail the given hypothesis. Rather than selecting a subset of hypotheses for fine-tuning, I train the DeBERTa model on the full set of 13 hypotheses simultaneously. Training on all hypotheses allows the model to learn the distinct entailment patterns associated with both factual claims (e.g., whether a country appears on a Watch List) and interpretive claims (e.g., whether the author believes a country has made efforts to combat IP violations). The training hyperparameters are set as follows: learning rate at 2×10^{-5} , weight decay of 0.03, and batch size of 8. The small learning rate and non-zero weight decay reduce the risk of overfitting to idiosyncratic patterns in the limited training data.

To evaluate out-of-sample performance, I implement a stratified 10-fold cross-validation procedure following the framework outlined in Grimmer et al. (2022). The labeled corpus of 300 paragraphs is randomly partitioned into 10 folds of roughly equal size, with stratification to preserve the label distribution across folds. For each fold $j = 1, \dots, 10$, the

model is trained on the remaining nine folds (approximately 270 paragraphs, yielding roughly 3,510 NLI pairs) and evaluated on the held-out fold (approximately 30 paragraphs). Crucially, the entire preprocessing pipeline—vocabulary construction, embedding computation, and model fitting—is executed within each fold using only the training partition, so that no information from the validation set leaks into the model. This “full pipeline inside each fold” approach is what Grimmer et al. (2022) emphasize as essential for honest validation in text-as-data applications, because computing embeddings or selecting features on the full sample would leak information from the test fold into training and inflate performance estimates. All reported performance metrics are therefore fully out-of-sample: every prediction is made on paragraphs the model has never seen during training.

Figure A.1 displays the aggregate confusion matrix from the 10-fold cross-validation. I dichotomize the continuous DeBERTa scores into negative (score ≤ 0) and positive (score ≥ 1) bins and compare them against the hand-coded labels. The model correctly identifies 167 of 224 negative paragraphs (74.6% recall) and 65 of 76 positive paragraphs (85.5% recall), yielding an overall accuracy of 77.3% and a weighted F1 score of 0.787. The asymmetry—higher recall for the positive class but lower precision (53.3%)—reflects the conservative threshold placement and the class imbalance in the labeled data. Importantly, the false negative rate for the positive class is low (14.5%), meaning that the model rarely misses genuinely positive IP assessments.

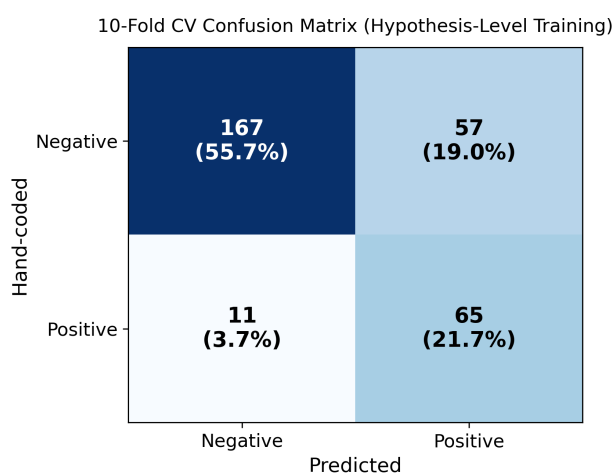


Figure A.1: Aggregate confusion matrix from stratified 10-fold cross-validation. Rows correspond to hand-coded labels and columns to model predictions. The model achieves an overall accuracy of 77.3% and a weighted F1 of 0.787.

Figure A.2 reports the per-hypothesis out-of-sample weighted F1 scores, which reveal a clear and intuitive pattern. Factual hypotheses (H1–H4, H12–H13, shown in blue)

achieve consistently high F1 scores ranging from 0.89 to 0.99, while interpretive hypotheses (H5–H11, shown in orange) yield moderately lower scores between 0.69 and 0.88. This distinction is expected. Factual hypotheses, such as whether a country is designated as the Priority Foreign Country or appears on the Watch List, have explicit textual markers that the model can readily identify, whereas interpretive hypotheses, such as whether the author believes a country has made efforts to combat IP violations, require more contextual, nuanced reading. The pooled weighted F1 across all 13 hypotheses is 0.870, indicating strong overall classification performance that is robust across hypothesis types.

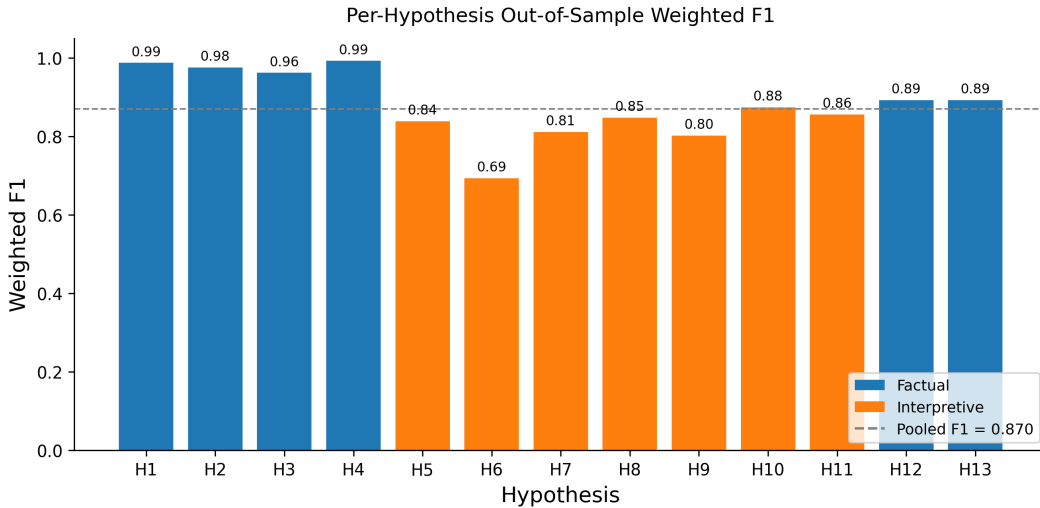


Figure A.2: Per-hypothesis out-of-sample weighted F1 scores from 10-fold cross-validation. Blue bars denote factual hypotheses; orange bars denote interpretive hypotheses. The dashed line marks the pooled F1 of 0.870.

Beyond hypothesis-level classification accuracy, I assess how well the model’s aggregate predicted scores track the hand-coded ground truth at the paragraph level. Figure A.3 plots the out-of-sample raw predicted score against the hand-coded score for each of the 300 paragraphs, with points colored by their cross-validation fold to illustrate that performance is stable across partitions. The Pearson correlation is $r = 0.676$ and the Spearman rank correlation is $\rho = 0.689$, both indicating a moderately strong positive association between predicted and hand-coded scores. The scatter around the best-fit line reflects the inherent difficulty of mapping nuanced diplomatic language onto a continuous scale, but the overall pattern confirms that the DeBERTa-based measure captures meaningful variation in IP evaluation sentiment across the full range of the score distribution.

After validating the model’s performance through this cross-validation exercise, I train the final model on the full set of 300 labeled paragraphs and apply it to the entire NTE

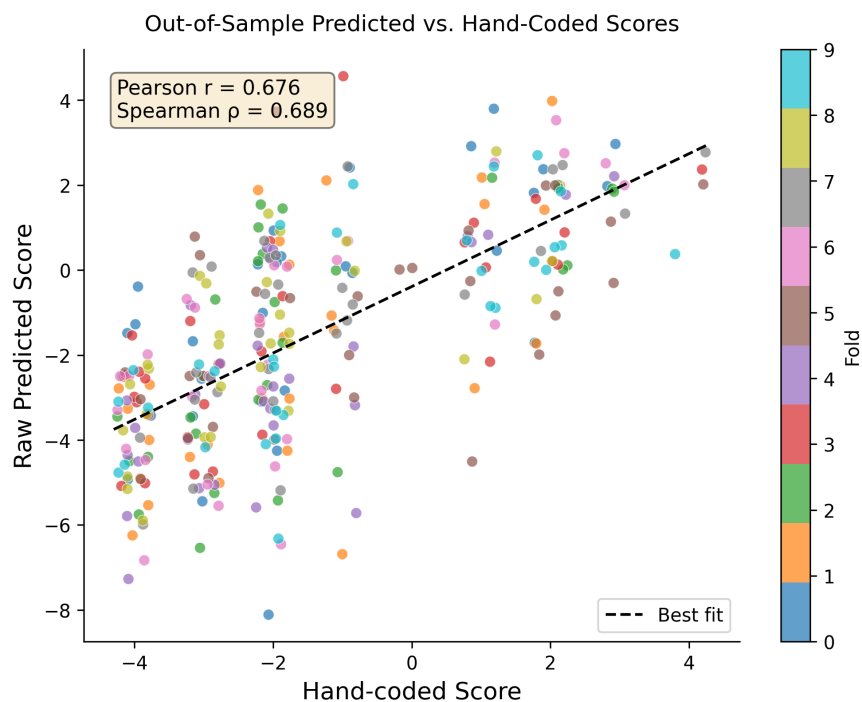


Figure A.3: Out-of-sample predicted scores versus hand-coded scores for 300 NTE IP paragraphs. Each point represents a paragraph, colored by cross-validation fold. The dashed line is the best-fit line. Pearson $r = 0.676$; Spearman $\rho = 0.689$.

corpus. Each IP paragraph is tested against all 13 hypotheses, and the resulting entailment probabilities are combined via the weighted sum specified in Table A.1. The weighted sums are then converted into a -5 to $+5$ scale to form the final country-year IP evaluation proxy.

A.2 Estimation results

Table A.2: Main results. YFE-only specification

	Aid $\times$ Democracy			IFC $\times$ Non-democracy		
	(1) Undiscl. info	(2) Patents	(3) Enforcement	(4) Undiscl. info	(5) Patents	(6) Enforcement
DeBERTa score _{<i>t</i>-1}	0.730** (0.100)	0.451* (0.222)	1.978* (0.735)	0.250* (0.102)	0.324* (0.122)	1.229* (0.453)
US aid _{<i>t</i>-1}	0.026 (0.022)	0.043 (0.045)	0.112 (0.146)			
IFC loan _{<i>t</i>-1}				0.026 (0.020)	0.024 (0.021)	0.095 (0.082)
DeBERTa _{<i>t</i>-1} $\times$ US aid _{<i>t</i>-1}	-0.034** (0.005)	-0.014 (0.011)	-0.072+ (0.037)			
DeBERTa _{<i>t</i>-1} $\times$ IFC loan _{<i>t</i>-1}				-0.016+ (0.009)	-0.015 (0.009)	-0.064+ (0.036)
Controls	✓	✓	✓	✓	✓	✓
Country FE	—	—	—	—	—	—
Year FE	✓	✓	✓	✓	✓	✓
<i>N</i>	739	739	739	314	314	314
R ²	0.418	0.422	0.434	0.432	0.412	0.435

Standard errors clustered at country level in parentheses
Signif. Codes: **: 0.01, *: 0.05, +: 0.1

A.3 Alternative identification. Two-way fixed effects and Mundlak decomposition

The main-text specifications in Table A.2 identify the interaction coefficients from cross-country variation in donor engagement, conditional on year fixed effects (YFE) and the full covariate set. Some readers may prefer to see how the interaction signals behave under a more demanding identification strategy that restricts attention to within-country year-over-year variation. Table A.3 reports the corresponding two-way fixed effects specifications for both subsamples, which add country fixed effects to the main design. Table A.4 then reports a Mundlak decomposition (Mundlak, 1978) that partitions the key interaction into its between-country and within-country components, so that both sources of identifying variation can be read off the same regression.

Table A.3: Alternative identification. Two-way fixed effects

	Aid $\times$ Democracy			IFC $\times$ Non-democracy		
	(1) Undiscl. info	(2) Patents	(3) Enforcement	(4) Undiscl. info	(5) Patents	(6) Enforcement
DeBERTa score _{<i>t</i>-1}	0.663** (0.078)	0.399* (0.160)	1.949** (0.580)	0.224 (0.133)	0.197+ (0.113)	0.818+ (0.470)
US aid _{<i>t</i>-1}	0.033 (0.020)	-0.006 (0.036)	0.040 (0.131)			
IFC loan _{<i>t</i>-1}				0.021+ (0.012)	0.016+ (0.009)	0.068+ (0.038)
DeBERTa _{<i>t</i>-1} $\times$ US aid _{<i>t</i>-1}	-0.034** (0.005)	-0.018+ (0.009)	-0.094** (0.031)			
DeBERTa _{<i>t</i>-1} $\times$ IFC loan _{<i>t</i>-1}				-0.009 (0.007)	-0.009 (0.006)	-0.036 (0.024)
Controls	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
<i>N</i>	738	738	738	335	335	335
R ²	0.610	0.727	0.708	0.678	0.821	0.785
Within R ²	0.133	0.124	0.116	0.095	0.105	0.105

Standard errors clustered at country level in parentheses

*Signif. Codes: **: 0.01, *: 0.05, +: 0.1*

Table A.3 shows that the interaction between the DeBERTa score and US aid in the democracy subsample remains negative and significant for undisclosed information under the two-way fixed effects design, while the interactions for patents and enforcement attenuate toward zero under within-country identification. The corresponding interaction between the DeBERTa score and IFC loan amount in the non-democracy subsample is negative in direction across all three TRIPS-plus dimensions but does not reach conventional significance

under the two-way fixed effects specification. This pattern, stronger in the cross-section than within-country, aligns with the theoretical mechanism being a between-country regime contrast rather than a within-country year-to-year adjustment, and it motivates the Mundlak decomposition reported next.

Table A.4: Identification robustness: Mundlak decomposition

	Aid $\times$ Democracy			IFC $\times$ Non-democracy		
	(1) Undiscl. info	(2) Patents	(3) Enforcement	(4) Undiscl. info	(5) Patents	(6) Enforcement
DeBERTa score $_{t-1}$	0.842** (0.154)	0.544+ (0.307)	2.360* (1.037)	0.305* (0.141)	0.358* (0.153)	1.391* (0.591)
Within: $x_{it} - \bar{x}_i$	0.009 (0.023)	-0.002 (0.039)	-0.002 (0.132)	0.021 (0.015)	0.015 (0.012)	0.063 (0.049)
Between: $\bar{x}_i$	0.026 (0.036)	0.066 (0.078)	0.157 (0.257)	0.030 (0.034)	0.028 (0.042)	0.107 (0.158)
DeBERTa $_{t-1} \times$ Within	-0.009 (0.014)	-0.001 (0.021)	-0.002 (0.080)	-0.008 (0.008)	-0.008 (0.008)	-0.033 (0.029)
DeBERTa $_{t-1} \times$ Between	-0.040** (0.009)	-0.019 (0.016)	-0.093 (0.056)	-0.020 (0.012)	-0.018 (0.013)	-0.077 (0.050)
Controls	✓	✓	✓	✓	✓	✓
Country FE	—	—	—	—	—	—
Year FE	✓	✓	✓	✓	✓	✓
N	739	739	739	339	339	339
R^2	0.422	0.423	0.435	0.434	0.414	0.439

Note: x denotes US aid (cols. 1–3) or IFC loan (cols. 4–6). Standard errors clustered at country level in parentheses. Signif. Codes: **: 0.01, *: 0.05, +: 0.1

Table A.4 makes the identifying variation explicit. For aid in the democracy subsample (columns 1–3), the between-country interaction (DeBERTa $_{t-1} \times \bar{x}_i$) is negative in all three specifications and strongly significant for undisclosed information, while the within-country interaction is indistinguishable from zero in every case. This pattern is theoretically clarifying. The compensation mechanism operates in the cross-section of country-level donor engagement rather than in year-to-year adjustments within a given country. For IFC lending in the non-democracy subsample (columns 4–6), both the between and within interactions are negative across all three outcomes, but neither reaches conventional significance. The between-country coefficients are larger in magnitude and hover near the 10 percent threshold, consistent with the cross-sectional pattern reported in Table A.2. The overall conclusion is that the interaction signals documented in Table A.2 are carried primarily by between-country variation, and that the two-way fixed effects specifications in Table A.3 test a more demanding estimand than the theoretical mechanism strictly requires.